

Measuring Image Dependence, Not Just Response Sensitivity: Controlled Ablation Tests on Brain-MRI Question Answering

Anonymous ACL submission

Abstract

Accuracy changes after removing or swapping an image do not by themselves show that a multimodal LLM uses the image correctly; they may reflect response sensitivity alone. We test image dependence by comparing ablation against the answer prior, scoring donor-image swaps against the donor key, and requiring an image-only positive control on brain-MRI questions from the LUMIERE glioblastoma cohort.

A reference model reads sequence and lesion side from the image (95.7% and 100.0%, versus 21.7% and 36.8% without it) and tracks donor swaps, showing the tasks are feasible. Four open-weight models show limited image dependence: only one image-effect test survives multiplicity correction, its accuracy remains below the most common key’s share, donor tracking is indistinguishable from random assignment, and none exceeds the best category lookup on the management rule. Two bf16 re-runs remain near chance on the control, making 4-bit serving an unlikely sole explanation.

A motivating audit finds that a slice-based enhancement rule matches the expert RANO rating for 38% of 197 determinate follow-ups, so those ratings are not treated as validated targets. We measure observable image-dependent behavior, not internal image use; keys are segmentation- or rule-derived and not clinician-validated.

1 Introduction

Medical multimodal LLMs are often evaluated by final-answer accuracy, which cannot distinguish image-based reasoning from reliance on textual cues, option patterns, or guessing. Such shortcuts are well documented: models exploit annotation artifacts in natural language inference (Gururangan et al., 2018), syntactic heuristics (McCoy et al., 2019), and unfaithful stated reasoning (Turpin et al., 2023). In chest-radiograph question answering, models can rely on linguistic patterns or irrelevant

image regions (Nguyen et al., 2025). Image ablation or swapping is informative only when the displayed image contains enough evidence to recover the answer.

We ask when an image intervention measures image dependence rather than merely response sensitivity. An accuracy difference between an image arm and a text-only arm is confounded by the answer prior (a no-image arm that scores below chance inflates it), and an answer that changes when the image is swapped need not move toward the answer the new image supports. We therefore read the ablation together with three additions: reference values for each phase (chance and the most common key); a matched donor swap scored against the donor’s key with the no-image prior subtracted; and an audit of whether the key can be recovered from what is displayed, so that answerability is checked rather than assumed. Protocols that perturb the text or the image and report the change in accuracy, such as HEAL-MedVQA (Nguyen et al., 2025), address the same concern; we did not run their protocol, so the comparison is by design, not measured, and Section 5 gives a worked case in which the ablation alone points to image use and the additions do not. We instantiate the questions on the five OmniBrainBench phases (Peng et al., 2026) as same-patient chains from the LUMIERE glioblastoma cohort (Suter et al., 2022): anatomical and imaging assessment (AIA), lesion identification and localization (LIL), diagnostic synthesis and causal reasoning (DSCR), prognostic judgment and risk forecasting (PJRF), and therapeutic cycle management (TCM); Figure 1 summarizes the design. The chain is an extension that lets us also test whether a model uses stated facts and upstream context in the management step. A motivating audit of the expert RANO ratings found that they cannot be treated as validated targets for the displayed slices (Section 3.1), which motivates an evaluation with explicit key prove-

nance and fixed controls (Sections 3.2–4).

The paper makes three contributions:

- A protocol that keeps three outcomes apart, accuracy benefit, output sensitivity and donor tracking, and reads image ablation against the answer prior, with a worked case (Section 5) in which the ablation alone credits Llama-4-Scout with image use on the diagnosis category although its accuracy with the image stays below the most common key’s share and its donor gain is not separable from a random donor assignment.
- An item set with an explicit origin for each key, built on LUMIERE, and a motivating audit of the expert labels: an enhancement-based rule reproduces the expert RANO category for 38% of the 197 follow-ups on which it is determinate (coverage is reported separately: it is undetermined for 44 of 241), so we do not treat those ratings as validated targets for the displayed slices; the audit does not show which side is wrong. The released OmniBrainBench groupings do not give same-patient chains (only 2 verifiable cross-phase links among 6,823 questions).
- An evaluation of four open-weight models and one reference model under a prespecified control suite. The reference model shows image-dependent performance on sequence and lesion-side questions, whereas the open models are limited and inconsistent; on the diagnosis-category task the most common key covers 58.7% of items, so accuracy there is hard to separate from a class prior even for the reference model. As an extension, the chain tests the use of stated facts and upstream context on the management step.

2 Related Work

Shortcuts and faithfulness. Models can reach correct labels without performing the intended reasoning. Hypothesis-only baselines exploit annotation artifacts in NLI datasets (Poliak et al., 2018; Gururangan et al., 2018), models adopt heuristics that fail on HANS (McCoy et al., 2019), and biased input features can alter chain-of-thought answers without appearing in the stated rationale (Turpin et al., 2023).

Answering without the image. Visual question-answering models often exploit language priors. Goyal et al. (2017) countered this by pairing questions with images yielding different answers, while Chen et al. (2024) show that many multimodal

benchmark items do not require visual content. Similar failures appear in medicine: Yan et al. (2025) report severe degradation under adversarial diagnostic counterparts, Jin et al. (2024) find flawed rationales in 35.5% of GPT-4V’s correct answers, and Nguyen et al. (2025) (HEAL-MedVQA) use textual and visual perturbations to localize shortcuts in chest-radiograph QA. We extend this logic to brain-MRI questions and add a donor swap scored against the donor’s key with the no-image prior subtracted, per-phase reference values, and a key-recoverability audit. Unlike correctness on complementary pairs (Goyal et al., 2017), which requires the model to answer each image correctly, this statistic measures movement toward what the swapped image supports and so is informative even for models that are rarely correct, as ours are; we did not run HEAL-MedVQA’s protocol, so that comparison is one of design. The same-patient chain extends the questions to interventions on supplied clinical facts. UniMod (Gu et al., 2026) addresses modality imbalance during training by supervising image-only, text-only, and fused predictions separately; we instead evaluate existing models for observable image dependence.

Brain benchmarks and longitudinal data. OmniBrainBench (Peng et al., 2026) contains 6,823 closed-ended brain-imaging questions spanning five clinical phases and 15 modalities. Its supplement describes a patient-level ADNI extension, but we did not evaluate it. LUMIERE (Suter et al., 2022) and MU-Glioma-Post (Mahmoud et al., 2025) provide longitudinal glioma imaging, treatment, and outcome data but no question layer.

3 Building Chains from LUMIERE

Why not OmniBrainBench. A chain needs one patient’s phases. We checked whether any image is shared across questions of different phases within a grouped case: of 6,823 closed-ended questions only 2 verifiable cross-phase patient links exist (both in VQA_RAD, one AIA and one LIL question each), and a second method based on filename identifiers agreed; the counts are lower bounds (Appendix A). Each sub-corpus was built for one task, so a patient is essentially asked one phase-type of question. This is an incompatibility between the released groupings and our evaluation, not a defect in a promise its authors made.

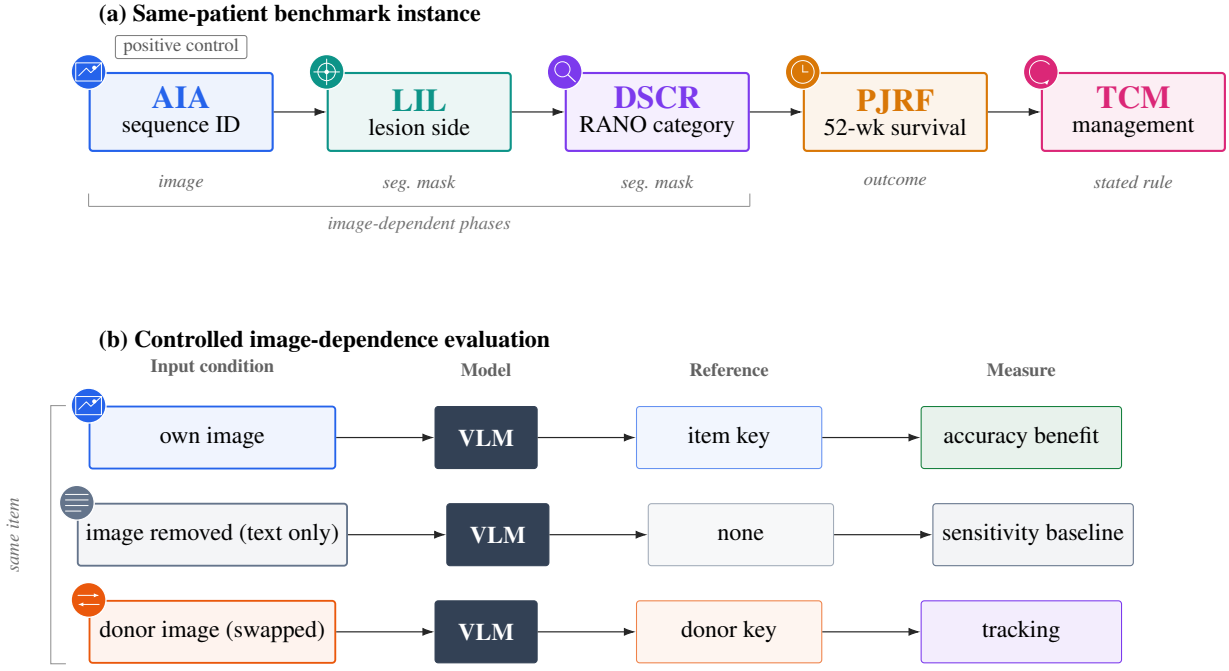


Figure 1: Evaluation overview. A stratified LUMIERE follow-up instantiates a five-phase same-patient benchmark chain. The image-dependent phases (AIA, LIL, DSCR) are each evaluated under three controlled input conditions: the patient’s own image, text only, and a donor-image swap. AIA serves as the positive control.

LUMIERE. LUMIERE (Suter et al., 2022) has 91 glioblastoma patients with longitudinal MRI (T1, contrast-enhanced T1, T2, FLAIR), automated segmentations (DeepBraTumIA and HD-GLIO-AUTO; Suter et al., 2022; Kickingreder et al., 2019), RANO ratings by one neuroradiologist (Wen et al., 2010), and overall survival for most patients. Segmentations are automated and were not manually verified; the ratings are expert-assigned.

3.1 Motivating audit: can an automated rule reproduce the expert ratings from the displayed slices?

The natural key for a diagnosis-category question is the expert RANO rating, but whether that rating can be recovered from the displayed slices is untested. We tested it with an automated reproducibility audit: for each first-line follow-up we tracked one enhancing lesion across scans (at each scan, the 3D-connected enhancing component nearest to the largest component of the post-operative baseline), measured it on its own largest axial cross-section, computed the RANO bidimensional product (longest diameter times longest perpendicular diameter) and applied the enhancement criteria (progression: measurable disease at least 25% above the smallest earlier measurement, the nadir,

or measurable disease where the nadir had none; response against the post-operative baseline; measurable means both diameters at least 10 mm). The rule has no new-lesion criterion beyond that. Coverage: of 241 first-line follow-ups (one per rated, imaged timepoint before any second surgery), 44 are undetermined because no measurable enhancing disease exists on the baseline or nadir scan, so the rule abstains on them. Agreement, our primary number, is computed over the 197 determinate follow-ups: the rule reproduces the expert category for 75 (38%, Wilson 95% interval 32 to 45%, and 29 to 48% when the 59 patients are resampled instead; counting the undetermined ones as failures would give 31%, which mixes abstention with disagreement). Most of the 122 disagreements are the rule returning complete response where the expert rated progressive or stable disease. Disagreement has several possible sources (segmentation or measurement error, a different target lesion or reference scan, our simplified rule, or evidence outside the displayed slices) that this audit does not separate; LUMIERE records that one neuroradiologist rated the studies but not what they used. Our automated product correlates only moderately with the rater’s recorded target-lesion diameters (Spearman 0.39 over 81 follow-ups), so measurement error is one plausible contributor, though the checkable subset

was too small to establish a size-closeness difference either way (Appendix D). The audit therefore shows these ratings cannot be treated as validated targets for the displayed inputs; it does not measure how many are unanswerable, which would need qualified reviewers on a sample of the disagreements.

3.2 The item set

The keys have four different origins (Table 1), and we make no uniform claim that each is a function of what the model sees: AIA is the sequence rendered; LIL and DSCR are derived from automated segmentation masks the model does not see (recoverable from the slices only if the segmentation is right; these are segmentation-derived targets reviewers have not validated); PJRF is the recorded outcome, scored probabilistically since a future outcome is not a function of the stated facts; and TCM is the output of a stated management rule, testing adherence rather than clinical correctness. All random choices use fixed seeds and never look at model output.

Cohort. We select one first-line follow-up per patient (rated and imaged before any second surgery, with an imaged post-operative reference scan), in strata defined by the TCM rule class and the DSCR category, with quotas and seed fixed in advance, giving 46 patients (counts by phase and key in Appendix D). Selection used tabular records and segmentation masks only, and progressive-disease follow-ups within 2 weeks of the 12-week window are left out.

Images. Each image is an atlas-space axial slice, skull-stripped by the dataset authors with HD-BET (Isensee et al., 2019), at 4 $\times$ magnification with a 20 mm scale bar, in radiological orientation (checked against the affine). Slice levels are those on which the key is computed.

AIA, LIL, DSCR. AIA is the positive control: its key is the sequence rendered, balanced over four sequences (the most common covers 26.1% of items), so text alone can reach only chance in expectation. LIL’s key is the hemisphere and anterior/posterior half of the largest tumor-core component on the displayed slice; lesions within 10 mm of a midline, smaller than 100 mm² or with under 80% of their pixels on their centroid’s side of the midline are not asked about. DSCR’s stem states the RANO enhancement criteria, and its key is the category

computed from the bidimensional products.

PJRF. The prediction time is the follow-up scan and the outcome is death within 52 weeks of it, from the LUMIERE survival record. The model picks one of four probability bins (midpoints .10, .30, .50, .80) and is scored with the Brier score, not by whether it picked a “correct” bin. The base rate (Table 2) reflects our selection, not a population rate. The origin and censoring of the survival time are undocumented (Appendix F), so PJRF is secondary and no other claim in this paper rests on it.

TCM. LUMIERE records no treatment after chemoradiotherapy, so a documented treatment is not available and the key is a rule-based recommendation, not what was done: non-progressive disease continues adjuvant temozolomide; progressive disease within 12 weeks of completing radiotherapy is confirmed with repeat MRI before any change; later progressive disease prompts a change of therapy. The base stems supply the clinical facts but not this rule; only a post hoc condition states the rule in the prompt (Section 5). The 12-week window is from the RANO criteria (Wen et al., 2010); because we cannot assess the radiation field, the rule is a simplification (see Limitations). Radiotherapy dates are not in LUMIERE, so the stem states the weeks since chemoradiotherapy under an assumed standard schedule (ending 10 weeks after surgery). The same diagnosis therefore maps to different actions depending on the stated timing (7 repeat-MRI and 20 change-therapy items among 27 progressive-disease items), which lets a lookup of the category alone be told apart from use of the timing fact.

4 Controls and Pre-specified Analysis

The analysis plan (hypotheses, margin, correction family and decision rules) was written before any model run; its hash is recorded in the project log, and the file is not externally timestamped. Models are the four open-weight models (MedGemma-4B (Sjellergren et al., 2025), Gemma-3-12B and Gemma-3-27B (Gemma Team, 2025), Llama-4-Scout; 4-bit, greedy decoding, 800 tokens; the reference model uses 16,384 tokens because its hidden reasoning counts against the cap and 800 would truncate; unparseable responses score incorrect). Each condition asks one question with a manufactured upstream context, so a phase’s effect is not

Phase	Question	Model-visible inputs	Key	<i>n</i>
■ AIA	Which MRI sequence is shown?	one axial slice (T1, T1 after contrast, T2 or FLAIR)	the sequence rendered (balanced)	46
■ LIL	Hemisphere and anterior/posterior half of the main tumor region	follow-up contrast-enhanced T1 slice at the level of the largest tumor region	position of the largest tumor-core component on that slice, relative to the midline and the brain's front-to-back midpoint	38
■ DSCR	RANO enhancement category at follow-up	baseline, follow-up and (when different from the baseline) nadir slices at the largest cross-section of the tracked enhancing lesion; the RANO thresholds are stated	category from the bidimensional products on those slices	46
■ PJRF	Probability of death within 52 weeks of this scan	age, sex, MGMT, IDH, extent of resection, weeks since surgery; alive at the scan	recorded outcome; scored by Brier score, not accuracy	45
■ TCM	Next management step	the same clinical facts plus weeks since chemoradiotherapy; the category comes from the chain	rule: non-PD continue; PD within 12 weeks of radiotherapy repeat MRI first; PD later change therapy	46

Table 1: The phases. Stems for AIA, LIL and DSCR contain no patient-specific fact and every option set contains every possible answer, so a swapped image never conflicts with the text and the donor’s answer is always available. DSCR shows two images, or three when the nadir differs from the baseline.

mixed with the model’s own earlier answers.

- **E1, image effect** (AIA, LIL, DSCR): accuracy with the image minus text only, paired by patient, with a patient-bootstrap interval. The margin is ± 10 percentage points, fixed in advance; “within margin” requires the whole interval inside it, which at these sample sizes is reachable only when few items differ between the arms. Holm correction over the 12 model-phase tests; other contrasts are descriptive and unadjusted, with exact McNemar tests (McNemar, 1947) and Wilson intervals (Wilson, 1927) where a proportion is reported. Because the bootstrap collapses to zero width when every pair is concordant, we also report, post hoc, an add-half interval for the paired difference (Agresti and Min, 2005) and the exact *p*. **AIA is the positive control:** the plan’s operational reading rule is that for a model whose unadjusted AIA interval is not above zero, the test has not been shown to detect image use, and E1 on LIL and DSCR supports no conclusion about it.
- **E2, donor tracking:** swap the image for a donor’s with a different key (one donor per patient and phase, same number of images); $G = 1[\text{swap} = \text{donor key}] - 1[\text{text only} = \text{donor key}]$, so the no-image prior is subtracted. The estimand is conditional on the fixed assignment, in which a patient donates to at most two recipients; donor-dependence and validation checks are post hoc (Appendix D).
- **E3, use of stated facts** (TCM): with the true con-

text, move the stated timing across the 12-week window (*flip_window*) or replace the stated category by one that maps to another action class (*flip_label*). The registered summary was the share of answers equal to the rule’s new answer; because that counts answers that had not changed, we instead report each item’s transition relative to the true-context answer (Table 10) and take moved over informative items as the rate of interest. A category lookup alone reaches at most 84.8% accuracy (56.5% if it always confirms first) and cannot follow the timing flip.

- **E4, upstream context:** true, randomly wrong (each earlier answer replaced by a seeded random incorrect option) or absent context.
- **E5, PJRF** (secondary; results in Appendix F): Brier score and skill against the leave-one-out base rate, and AUC.

We report three observable outcomes separately: accuracy benefit (E1), output sensitivity (the share of items whose answer letter differs between the image and text-only arms, and between the swapped and own image), and correct donor tracking (E2); none observes internal processing. Output sensitivity, the transition tables and the output-handling counts are post hoc, descriptive analyses. The ordinary five-phase chain (own answers as context) is run with and without images as a secondary analysis; each arm regenerates its upstream answers, so later-phase differences are total effects through the chain.

Phase	n	Chance	Most common key
AIA	46	25.0	26.1%
LIL	38	25.0	28.9%
DSCR	46	25.0	58.7%
TCM	46	25.0	43.5% (lookup $\leq$ 84.8%)
PJRF	45	(Brier)	base rate 62.2%

Table 2: Non-model references on the items: chance, share of the most common key, the ceiling of a category-only TCM lookup, and the PJRF base rate.

Reference and precision controls. Three of the four open models did not meet the operational AIA gate (Section 5), so we added three post hoc controls, each reported apart from the four-model Holm family and run under the same prompts, temperature 0 and decision rules (tags and digests in Appendix F). (i) A reference (positive-control) model, Gemini-3.6-Flash (Google API), on E1 and E2; empty API responses are counted and rerun (0 of 386 records), and, being an API model, it can change without notice. (ii) A precision control, Gemma-3-27B rerun in bf16 (vLLM, one GPU) on the same conditions; its reading rule, fixed in advance, was that an AIA interval above zero and clearly above the 4-bit model’s would attribute part of the gap to quantization and a near-chance result would not, and the serving stack differs from the Ollama runs, so precision and stack are confounded (we draw no conclusion). (iii) A non-Gemma control, Pixtral-12B (bf16, vLLM), on E1 and E2 for AIA and LIL; its DSCR image conditions are excluded post hoc because the three-image prompt returned empty output for many records (Appendix F).

5 Results

5.1 Reference values without a model

Table 2 gives what constants and lookups reach. AIA is close to uniform over four answers, DSCR’s most common category is progressive disease for 58.7% of items, and a category-only TCM lookup reaches at most 84.8%.

5.2 Image controls and stated facts

The reference model reads the image. Gemini-3.6-Flash identifies the sequence for 95.7% of patients with the image and 21.7% without it (difference +73.9 points, 95% CI [+58.7, +87.0]), and it places the lesion for 100.0% against 36.8%. With the image swapped for a donor’s it gives the donor’s answer for 95.7% (AIA) and 100.0% (LIL) of pa-

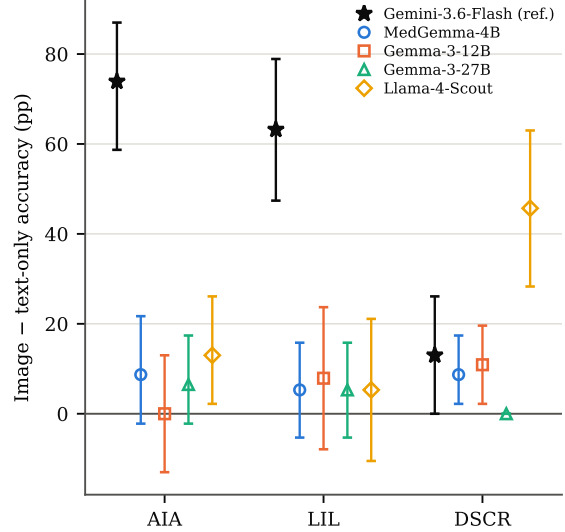


Figure 2: E1 image effect (image minus text-only accuracy, percentage points) per model by phase, with 95% bootstrap intervals; the same values as Table 3 (reference model from Table 7). The Gemini-3.6-Flash reference separates from the text-only arm on sequence and lesion side, whereas the four open models sit near zero, apart from Llama-4-Scout on the class-prior-confounded DSCR task. Precision/stack controls are omitted for legibility and reported in the text.

tients (Table 7). A model that reads the slices can therefore answer these items, which shows the tasks are feasible; it does not show that every item is valid, nor why a weaker model fails (perception, input handling or training). DSCR is different: the reference model scores 71.7% with the image and 58.7% without, where the most common key covers 58.7% of items, so most of its text-only accuracy is the class prior, and it recovers the key for only 71.7% of items. We therefore cannot establish that DSCR is answerable from the slices for every item (the shortfall may be perception, segmentation-derived keys the model does not see, or key error, which we cannot separate), so a null DSCR effect for another model cannot be read as a failure to use the image.

Open models: image effect (E1). Sequence identification stays between 26.1 and 37.0% (chance 25%), and Table 3 gives every cell; Figure 2 plots the per-model effect with its interval. The operational AIA gate (an unadjusted interval above zero) is met only by Llama-4-Scout, whose gain is small beside the reference model’s; MedGemma-4B, Gemma-3-12B and Gemma-3-27B fall below it. The gate is a conservative screen,

Model	Phase	<i>n</i>	Image	Text-only	Δ [95% CI]	Add-half CI	Exact <i>p</i>	Answers differ	Interval
MedGemma-4B	AIA	46	34.8	26.1	+8.7 [−2.2, +21.7]	[−3.7, +20.4]	.289	30.4	○ inconclusive
MedGemma-4B	LIL	38	23.7	18.4	+5.3 [−5.3, +15.8]	[−5.8, +15.8]	.625	18.4	○ inconclusive
MedGemma-4B	DSCR	46	19.6	10.9	+8.7 [+2.2, +17.4]	[−0.5, +17.2]	.125	19.0	• CI above 0
Gemma-3-12B	AIA	46	26.1	26.1	+0.0 [−13.0, +13.0]	[−12.2, +12.2]	1.00	32.6	○ inconclusive
Gemma-3-12B	LIL	38	31.6	23.7	+7.9 [−7.9, +23.7]	[−9.3, +24.3]	.549	47.4	○ inconclusive
Gemma-3-12B	DSCR	46	58.7	47.8	+10.9 [+2.2, +19.6]	[+0.9, +20.0]	.062	11.9	• CI above 0
Gemma-3-27B	AIA	46	32.6	26.1	+6.5 [−2.2, +17.4]	[−3.6, +16.1]	.375	26.1	○ inconclusive
Gemma-3-27B	LIL	38	23.7	18.4	+5.3 [−5.3, +15.8]	[−5.8, +15.8]	.625	34.2	○ inconclusive
Gemma-3-27B	DSCR	46	58.7	58.7	+0.0 [+0.0, +0.0]	[−4.1, +4.1]	1.00	0.0	■ within ±10 pp
Llama-4-Scout	AIA	46	37.0	23.9	+13.0 [+2.2, +26.1]	[+0.8, +24.2]	.070	30.4	• CI above 0
Llama-4-Scout	LIL	38	39.5	34.2	+5.3 [−10.5, +21.1]	[−11.2, +21.2]	.754	44.7	○ inconclusive
Llama-4-Scout	DSCR	46	54.3	8.7	+45.7 [+28.3, +63.0]	[+26.0, +61.5]	<.001	88.1	• CI above 0

Table 3: E1: accuracy (%) with the image and with text only, with no upstream context, the paired difference in percentage points with its bootstrap interval and an add-half sparse-data interval (Add-half CI), the exact McNemar *p* (unadjusted), the share of items whose answer letter differs between the two arms (answers differ), and the interval reading (unadjusted; only Llama-4-Scout on DSCR survives Holm correction, and “CI above 0” is not otherwise a confirmatory result). The answers-differ share is over items present in all three arms (AIA 46, LIL 38, DSCR 42 of 46); accuracy is over all items in each phase (AIA 46, LIL 38, DSCR 46). AIA is the positive control.

not a claim that lesion or category reading requires sequence identification, so we read their LIL and DSCR effects with caution rather than as image use; it is near its boundary, as Llama-4-Scout meets it on the bootstrap interval alone (exact McNemar $p = .070$; none passes under the exact test). One image-effect test survives Holm correction, Llama-4-Scout on DSCR, but it does not show that the model reads the category from the slices: its accuracy with the image stays below the most common key’s share (58.7%), and its text-only accuracy is far below that share, so the difference does not establish that the image exceeds the prior (the two accuracies give the gap, not its mechanism). The DSCR intervals above zero for MedGemma-4B and Gemma-3-12B do not survive correction and are not read as image use; on LIL every interval includes zero. A test that does not survive correction is not evidence of no effect: every open-model AIA and LIL interval is wide and none lies inside the ± 10 -point margin. Each open model’s AIA answers concentrate on one sequence (its most frequent answer covers 52 to 76% of the 46 items), the pattern a fixed default with weak image use would produce; we cannot separate that from a failure to perceive the sequence, though every image reaches the model in each pipeline (Appendix F). Accuracy benefit is small, but answers are not insensitive to the image: the image and text-only arms differ for a substantial share of items in every phase (Table 3).

Precision control. Rerun in bf16, Gemma-3-27B identifies the sequence for 26.1% with the image and 26.1% without (against 32.6% at 4-bit; Table 7): the interval includes zero, so the rerun did not restore AIA performance, and it is indistin-

guishable from text-only on LIL and DSCR. Precision and serving stack changed together, so we draw no conclusion about quantization. A different-family bf16/vLLM model, Pixtral-12B, is likewise near chance on sequence (30.4% against 26.1%) and lesion side, with null donor tracking (its DSCR image arms excluded; Appendix F). That a non-Gemma bf16 model on a different stack also sits near chance on AIA makes 4-bit Ollama serving an unlikely sole cause, though scale and training still differ (full intervals and donor gains in appendix E).

Open models: donor tracking (E2). Swapping the image changes a substantial share of AIA and LIL answers (Table 6), but the new answer follows the donor’s key with an interval above zero in only two cells: MedGemma-4B on AIA (gain +10.9; exact $p = .062$) and Llama-4-Scout on DSCR (+28.6; $p = .029$), the phase in which its accuracy gain also survives correction; under an exploratory Holm correction over the twelve E2 cells (outside the registered family) the two become .688 and .348. Both survive resampling by donor and dropping any one donor, but neither is separable from a random assignment of donor keys (permutation $p = .059$ and .244; Appendix D), so for DSCR we cannot tell donor-specific tracking from a shift that any swapped image would produce, and this holds across re-drawn valid donor assignments while the reference model stays significant throughout (Appendix D). Every other interval includes zero, including both larger models on sequence and all four on lesion side. On sequence and lesion side the open-model gains are far below the reference model’s (+67.4 and +86.8; Table 7).

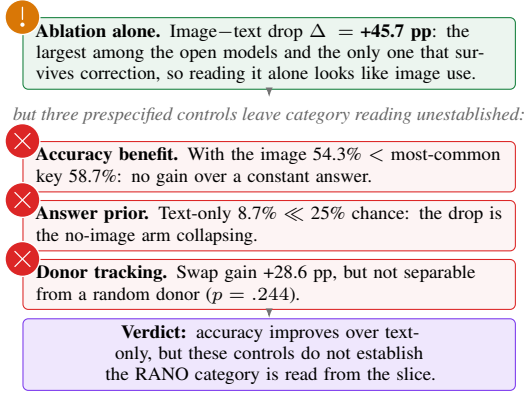


Figure 3: Worked example (Llama-4-Scout, DSCR): the ablation drop is the largest among the open models and survives correction, so reading it alone would credit image use, but the three prespecified additions (accuracy benefit over the answer prior, the prior itself, and donor tracking) do not establish that the category is read from the slice. The donor-dependence permutation behind the third point is a post hoc check.

Worked example: ablation without tracking.

Llama-4-Scout on DSCR (Figure 3) is the case in which reading the ablation alone would credit image use: its image-minus-text difference is the largest among the open models and the only one that survives correction (+45.7 [+28.3, +63.0], adjusted $p < .01$). The three prespecified additions do not bear out that reading: the image arm does not beat the most common key, the text-only arm scores far below the answer prior, and the donor gain is not separable from a random assignment ($p = .244$). A text-only ablation drop alone, the largest among the open models here, does not separate correct use of the slice from a collapse of the no-image arm; reading it against the answer prior and scoring the swap against the donor’s key does. Prior image interventions differ in construction rather than sharing this reading: HEAL-MedVQA replaces a queried region with the same region from a differently-labelled image (Nguyen et al., 2025), and balanced VQA scores correctness on complementary image pairs (Goyal et al., 2017). The ablation here shows response sensitivity to the image; these controls do not establish that the category is read from the slices.

Stated facts and context (E3, E4; extension).

With the true context, TCM accuracy is below the best category-only lookup (84.8%) for all four open models (Gemma-3-12B and Gemma-3-27B exceed the confirm-first lookup, 56.5%). Because accuracy alone does not show which facts a model uses,

we read each item against itself (Tables 9 and 10): replacing the stated category moves the answer for a minority of informative items in three models and a slim majority in Gemma-3-27B (25 of 43), and crossing the 12-week window moves it most for Llama-4-Scout and not at all for Gemma-3-27B; MedGemma-4B answers the most common class almost regardless, so its rule-following is inflated by answers already at the target. True context beats no context for three models and beats random wrong context for all four on TCM (unadjusted), but does not help on LIL or DSCR. Stating the rule in the prompt (post hoc) raises TCM accuracy for three models toward, but not above, the best lookup; part of Llama-4-Scout’s gain is better-formed output (Appendix F).

Chain (secondary). The ordinary chain (own answers as context) matters only for TCM, through the stated facts above; most image-minus-text differences are small with intervals including zero (uncorrected), the exception being Llama-4-Scout on AIA, which repeats its E1 result. We do not claim a causal-degradation effect across phases.

6 Conclusion

Accuracy on an expert-labeled brain-MRI chain cannot show whether a model reads the scan when the labels are not validated targets for what is displayed: an automated rule reproduces the expert RANO rating for 38% of the determinate follow-ups, and the benchmark we started from did not establish same-patient chains. On a LUMIERE chain with a stated origin for each key and a control suite fixed in advance, a reference model reads sequence and lesion side, whereas four open-weight models show limited and inconsistent image dependence: one accuracy gain survives correction (and stays below the most common key’s share), donor tracking is not separable from a random donor in any case, and none exceeds the best category lookup on the management rule. These results do not identify whether the models use the image internally; scale, training, serving stack and quantization are not separated, and clinician review of the keys and a second reference model remain outstanding.

Limitations

Clinician review is outstanding. No clinician has verified that each question is answerable from the displayed inputs or that each key is correct. LIL

and DSCR are derived from DeepBraTumIA segmentation masks unseen by the model, so segmentation errors propagate directly to these targets; the reference model supports task feasibility, not item-level validity. The enhancement rule further omits T2/FLAIR change, clinical status, and steroid dose, abstains when no measurable enhancing disease exists (44 of 241 follow-ups), and matches the expert category for 38% of determinate cases. Disagreement may reflect segmentation or measurement error, a different target lesion or reference scan, simplification of the rule, or evidence outside the displayed slices, which this audit cannot distinguish.

Our complete-response label is also weaker than clinical RANO: it is assigned whenever follow-up enhancement falls below the measurability threshold, potentially including residual sub-threshold disease, rather than requiring sustained disappearance of all enhancing and nonmeasurable disease. The rule tracks only the enhancing component nearest the baseline target lesion and does not implement RANO’s new-lesion criterion, under which an unequivocal new lesion implies progressive disease. Among the 46 follow-ups labeled complete response by the rule but progressive disease by the expert, 22 contain a distinct newly measurable enhancing component elsewhere, 8 contain a similarly sized component already present at the nadir, and 16 contain no other measurable enhancing component. Thus, the tracking rule explains some, but not all, disagreement. The expert rating is recorded for audit purposes but is not used as the benchmark key.

The management key is a rule, not a documented treatment. LUMIERE records no treatment after chemoradiotherapy. The TCM key therefore assumes a radiotherapy schedule ending 10 weeks after surgery and simplifies the 12-week RANO window, which can still permit progression when new enhancement lies outside the radiation field. PJRF further assumes that survival time and scan weeks share the surgery-week origin, while censoring is undocumented. With only 45 patients, a leave-one-out logistic model using the stated facts did not beat the base rate (Brier 0.283 vs. 0.246). This does not imply that the facts lack predictive information: the base rate reflects our stratified sample, and the outcome construction itself is unverified. We therefore treat PJRF as secondary and make no substantive claim from it.

Power and scope. With 38 to 46 patients per

phase, the pre-specified ± 10 -point margin is attainable only when few items differ between arms, so several comparisons remain inconclusive. Only one E1 image-effect test (Llama-4-Scout on DSCR) survives Holm correction. Moreover, MedGemma-4B, Gemma-3-12B, and Gemma-3-27B fail the preregistered AIA positive-control gate; under the analysis plan, their LIL and DSCR effects therefore support no conclusion. AIA tests only sequence identification, so passing it does not establish lesion reading. Donor images are from different patients and thus differ in anatomy beyond the manipulated feature, while text-only stems still refer to images.

The study uses a single-center, deliberately stratified cohort. The four open-weight models were served at 4-bit through Ollama, whereas the reference model is an API system of unknown size and precision, so between-model differences confound scale, training, quantization, and serving stack and cannot be attributed specifically to shortcut learning. bf16 reruns of Gemma-3-27B and a different-family model, Pixtral-12B, through vLLM also remained near chance on AIA, making 4-bit Ollama serving an unlikely sole explanation but not isolating quantization from serving, scale, or training; only these two models were rerun. The reference model was evaluated once at temperature 0, and API models may change over time.

Input handling was checked only in part. Post hoc checks confirmed that every image increased prompt token count, including the second and third DSCR images, and that each pipeline answered known-answer probes correctly (Appendix F). We did not verify that each serving stack passes the same image representation to its encoder as a reference implementation, and a single-patient check cannot establish preservation of MRI contrast across encoders. Thus, near-chance sequence identification in several open models remains unexplained: dropped images appear unlikely, but weak visual features or default-answer behavior remain possible. MedGemma-4B produced invalid JSON for 105 of 965 records and Llama-4-Scout for 46 of 965; pattern matching recovered the answer letter except for 0 MedGemma records (Appendix F). Intervals treat patients as exchangeable.

Ethics Statement

This work evaluates the diagnostic reasoning of general-purpose and medical-domain language models on real, de-identified patient imaging data;

it does not deploy, recommend, or endorse any model for clinical use, and no results in this paper should be interpreted as a clinical validation of any evaluated system. LUMIERE is used under its published non-commercial research license, and all patient data is accessed in its already de-identified, publicly released form; no new identifiable patient information is collected. The LUMIERE authors report that the cantonal ethics committee of Bern approved the studies and waived written informed consent, that the images are skull-stripped, and that dates are given as weeks relative to the pre-operative scan; we did not attempt to re-identify patients. An AI coding assistant (Claude Code) helped write and run the analysis code and edit the text; the authors are responsible for all content.

References

Alan Agresti and Yongyi Min. 2005. [Simple improved confidence intervals for comparing matched proportions](#). *Statistics in Medicine*, 24(5):729–740.

Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, and Feng Zhao. 2024. Are we on the right way for evaluating large vision-language models? In *Advances in Neural Information Processing Systems*, volume 37.

Gemma Team. 2025. [Gemma 3 technical report](#). *Preprint*, arXiv:2503.19786. Cited for model identification only.

Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2017. Making the V in VQA matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 6325–6334.

Zijian Gu, Weikai Lin, Shuang Zhou, Zihan Chen, and Song Wang. 2026. [UniMod: Enhancing multi-modal medical diagnosis through cross-modality and within-modality alignment](#). In *Proceedings of the 34th ACM International Conference on Multimedia (MM '26)*, Rio de Janeiro, Brazil. To appear; accepted, proceedings entry not yet verified.

Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. [Annotation artifacts in natural language inference data](#). In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, pages 107–112, New Orleans, Louisiana. Association for Computational Linguistics.

Fabian Isensee, Marianne Schell, Irada Pflueger, Gianluca Brugnara, David Bonekamp, Ulf Neuberger,

Antje Wick, Heinz-Peter Schlemmer, Sabine Heiland, Wolfgang Wick, Martin Bendszus, Klaus H. Maier-Hein, and Philipp Kickingereder. 2019. [Automated brain extraction of multisequence MRI using artificial neural networks](#). *Human Brain Mapping*, 40(17):4952–4964.

Qiao Jin, Fangyuan Chen, Yiliang Zhou, Ziyang Xu, Justin M. Cheung, Robert Chen, Ronald M. Summers, Justin F. Rousseau, Peiyun Ni, Marc J. Landsman, Sally L. Baxter, Subhi J. Al’Aref, Yijia Li, Alex Chen, Josef A. Brejt, Michael F. Chiang, Yifan Peng, and Zhiyong Lu. 2024. [Hidden flaws behind expert-level accuracy of multimodal GPT-4 vision in medicine](#). *npj Digital Medicine*, 7:190.

Philipp Kickingereder, Fabian Isensee, Irada Tursunova, Jens Petersen, Ulf Neuberger, David Bonekamp, Gianluca Brugnara, Marianne Schell, Tobias Kessler, Martha Foltyn, Inga Harting, Felix Sahm, Marcel Prager, Martha Nowosielski, Antje Wick, Marco Nolden, Alexander Radbruch, Jürgen Debus, Heinz-Peter Schlemmer, and 8 others. 2019. [Automated quantitative tumour response assessment of MRI in neuro-oncology with artificial neural networks: a multicentre, retrospective study](#). *The Lancet Oncology*, 20(5):728–740.

E. Mahmoud, J. Gass, Y. Dhemes, J. Greaser, K. Pogorzelski, E. Isufi, F. Garrett, J. Thacker, N. H. Tahon, J. Sinclair, L. Layfield, T. Altes, and A. Nada. 2025. [MU-Glioma Post: A comprehensive dataset of automated MR multi-sequence segmentation and clinical features](#). *Scientific Data*, 12:1847.

R. Thomas McCoy, Ellie Pavlick, and Tal Linzen. 2019. [Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3428–3448, Florence, Italy. Association for Computational Linguistics.

Quinn McNemar. 1947. [Note on the sampling error of the difference between correlated proportions or percentages](#). *Psychometrika*, 12(2):153–157.

Dung Nguyen, Minh Khoi Ho, Huy Ta, Thanh Tam Nguyen, Qi Chen, Kumar Rav, Quy Duong Dang, Satwik Ramchandre, Son Lam Phung, Zhibin Liao, Minh-Son To, Johan Verjans, Phi Le Nguyen, and Vu Minh Hieu Phan. 2025. [Localizing before answering: A benchmark for grounded medical visual question answering](#). In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, pages 7670–7678.

Zhihao Peng, Cheng Wang, Shengyuan Liu, Zhiying Liang, Zanting Ye, Minjie Ju, Peter Y. M. Woo, and Yixuan Yuan. 2026. [OmniBrainBench: A comprehensive multimodal benchmark for brain imaging analysis across multi-stage clinical tasks](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 42732–42743.

Adam Poliak, Jason Naradowsky, Aparajita Haldar, Rachel Rudinger, and Benjamin Van Durme. 2018. [Hypothesis only baselines in natural language inference](#). In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics*, pages 180–191, New Orleans, Louisiana. Association for Computational Linguistics.

Andrew Sellergren, Sahar Kazemzadeh, Tiam Jaroensri, et al. 2025. [MedGemma technical report](#). *Preprint*, arXiv:2507.05201. Cited for model identification only.

Yannick Suter, Urspeter Knecht, Waldo Valenzuela, Michelle Notter, Ekkehard Hower, Philippe Schucht, Roland Wiest, and Mauricio Reyes. 2022. [The LUMIERE dataset: Longitudinal glioblastoma MRI with expert RANO evaluation](#). *Scientific Data*, 9:768.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. 2023. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems*, volume 36.

Patrick Y. Wen, David R. Macdonald, David A. Reardon, Timothy F. Cloughesy, A. Gregory Sorensen, Evan-thia Galanis, John DeGroot, Wolfgang Wick, Mark R. Gilbert, Andrew B. Lassman, et al. 2010. [Updated response assessment criteria for high-grade gliomas: Response Assessment in Neuro-Oncology Working Group](#). *Journal of Clinical Oncology*, 28(11):1963–1972.

Edwin B. Wilson. 1927. [Probable inference, the law of succession, and statistical inference](#). *Journal of the American Statistical Association*, 22(158):209–212.

Qianqi Yan, Xuehai He, Xiang Yue, and Xin Eric Wang. 2025. [Worse than random? an embarrassingly simple probing evaluation of large multimodal models in medical VQA](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 19188–19205, Vienna, Austria. Association for Computational Linguistics.

A OmniBrainBench Compatibility Details

A chain presupposes that a case’s phases are one patient’s trajectory. In the 6,823-question closed-ended subset of OmniBrainBench (Peng et al., 2026) only 2 verifiable cross-phase patient links exist (both in VQA_RAD, one AIA and one LIL question); stripping modality, view and slice suffixes from filenames gave the same result. The release has no complete patient identifier, so these are lower bounds. Each sub-corpus was built for one clinical task, so a patient is essentially asked one phase-type of question, and reading source_file as patient identity would stitch together different

patients. This is an incompatibility with our evaluation, not a defect in the benchmark. The supplement’s patient-level ADNI extension is not a brain-tumor cohort and, as of September 2026, has label-generation scripts only; we did not evaluate it. We found no dataset with same-patient, multi-phase continuity and a question layer (Section 2), which motivates LUMIERE.

B Artifacts, Licenses, Compute and Intended Use

Data. LUMIERE (Suter et al., 2022) (91 patients, from Figshare) may be used only for non-commercial purposes, which our research use respects; its authors report ethics approval and a consent waiver by the cantonal ethics committee of Bern, skull-stripped images and dates blurred to weeks. We share the rendered slices the models saw, with attribution and for non-commercial research only, not the imaging archive. OmniBrainBench (CC BY-SA 3.0 per its dataset card) was used only to count cross-phase links (Appendix A) and is not redistributed.

Reproducibility artifact. An anonymous repository (link in the submission) holds the items with keys and key origins, the donor assignments, the exact prompt of every single-question condition, every model response used, a manifest of which run supplied each record and which later records replaced earlier ones, the analysis code with fixed seeds, and one command that regenerates every statistic and table and checks them against this paper. It excludes model weights, API keys and the full LUMIERE archive.

Models and terms. MedGemma-4B (Health AI Developer Foundations terms), Gemma-3-12B and Gemma-3-27B (Gemma Terms of Use), Llama-4-Scout (Llama 4 Community License) and Gemini-3.6-Flash (Google’s API terms) were used for research evaluation only, as these terms permit; no output is used clinically. Tags, digests and quantization are in Table 8. The MedGemma and Gemma 3 technical reports are cited only to identify those models; they are not peer-reviewed, and no claim rests on them. Llama-4-Scout has no technical report.

Software and compute. Ollama 0.33.0 served 4-bit Q4_K_M builds of the four open models (4.3, 12.2, 27.4 and 108.6 billion parameters; Llama-4-Scout is a mixture of experts with 17B active)

and vLLM 0.27.1 (PyTorch 2.13.0, Transformers 5.15.1) the two bf16 runs (Gemma-3-27B and Pixtral-12B); the analysis used Python 3.12.5 with NumPy 2.3.5, SciPy 1.18.1, pandas 3.0.5 and Pillow 12.3.0. Almost all GPU time was on one NVIDIA RTX PRO 6000 Blackwell GPU per job: about 154 GPU-hours for all LUMIERE work including failed and exploratory jobs, about 56 for the runs and input checks reported here. The Gemini runs made 525 API calls and about 1.1 million tokens including hidden reasoning tokens. Decoding was greedy with the caps in Section 4; no hyperparameters were tuned and each condition ran once.

Intended use. The items, code and results are for evaluation research only: not a clinical benchmark, keys not clinician-validated, and nothing here supports using a model for diagnosis, prognosis or treatment.

C A Worked Example

The patient is Patient-007, the first in id order with all five phases and a three-image DSCR item (a rule fixed before any answer was inspected). Figure 4 shows the images the models see and the AIA and LIL donor images used in the swap (the DSCR donor slices are omitted for space); each item is four-way multiple choice, with the full stems and options in the reproducibility artifact.

AIA (positive control). Which MRI sequence does this image show? *Key:* D) T1-weighted, after gadolinium contrast.

LIL. Where is the main tumor region (enhancing tumor, necrosis or resection cavity, not the surrounding edema) on this slice? *Key:* A) Right hemisphere, posterior half.

DSCR. What is the response category at follow-up? *Key:* B) Progressive disease.

The key comes from the bidimensional products of the tracked lesion on the three slices (10, 0 and 385 mm² at baseline, nadir and follow-up); there is no measurable enhancing disease at the nadir and a measurable lesion at follow-up (a new measurable lesion), which the rule calls progressive disease. The expert rating is PD (“Target L.: 1, 16mm x 19mm”), the same category, so this is not an audit disagreement.

TCM. Using your earlier assessment of this patient, what is the most appropriate next step? *Key:*

	AIA	LIL	DSCR
Key/donor	T1c/FLAIR	R post/L post	PD/PR
Gemini-3.6-Flash	T1c/T1c/FLAIR	R post/R ant/L post	PD/PD/CR
Llama-4-Scout	T1c/T1c/T1	L post/L post/L post	PD/SD/PD
Gemma-3-27B	T2/T1c/FLAIR	L ant/L ant/L post	PD/PD/PD
Gemma-3-12B	T2/T2/FLAIR	R ant/R ant/L post	PD/PD/PD
MedGemma-4B	T1/T1/T1	R ant/R ant/L ant	CR/CR/CR

Table 4: Answers for Patient-007 (own/text/swap in each cell), with the key and donor key; Gemini-3.6-Flash is the reference model. Labels: T1c is T1 after contrast; L/R hemisphere, ant/post half; PD, SD, PR, CR are RANO categories.

C) Change therapy now: bring re-resection, second-line treatment or a clinical trial to tumor board.

The base stem states the patient facts (see the artifact) but not the management rule; the rule is supplied in the prompt only in the post hoc explicit-rule condition (Section 5). The key follows that rule: non-PD → continue; PD and ≤ 12 weeks after radiotherapy → repeat MRI first; PD and later → change therapy (RANO 2010; Stupp schedule assumed). Here the category is progressive disease and chemoradiotherapy ended 53 weeks before the scan, so the class is escalate; changing the stated timing across the 12-week window or the stated category changes the rule’s answer (the fact flips of E3).

PJRF (secondary). Same patient facts as TCM; the model forecasts the probability of death within 52 weeks of the scan (recorded outcome: death), scored by the Brier score.

Donor swap. Each image is replaced by a donor’s whose key differs (AIA Patient-054, FLAIR not T1c; LIL Patient-023, L post not R post; DSCR Patient-019, PR not PD); the stem and options are unchanged, so the donor’s answer is always an option.

D Item Set Summary and Donor-Dependence Checks

Item set. From 59 patients with a first-line follow-up we keep 46. AIA has 46 items (T1 12, T1 after contrast 12, T2 11, FLAIR 11). LIL has 38: the rest have no tumor core, a core within 10 mm of a midline or crossing it, or one under 100 mm² (posterior left 11, posterior right 11, anterior right 9, anterior left 7). DSCR has 46 (progressive disease 27, complete response 9, stable disease 6, partial response 4), the key equalling the expert rating for 26. TCM has 46 (continue 19, repeat MRI first 7,

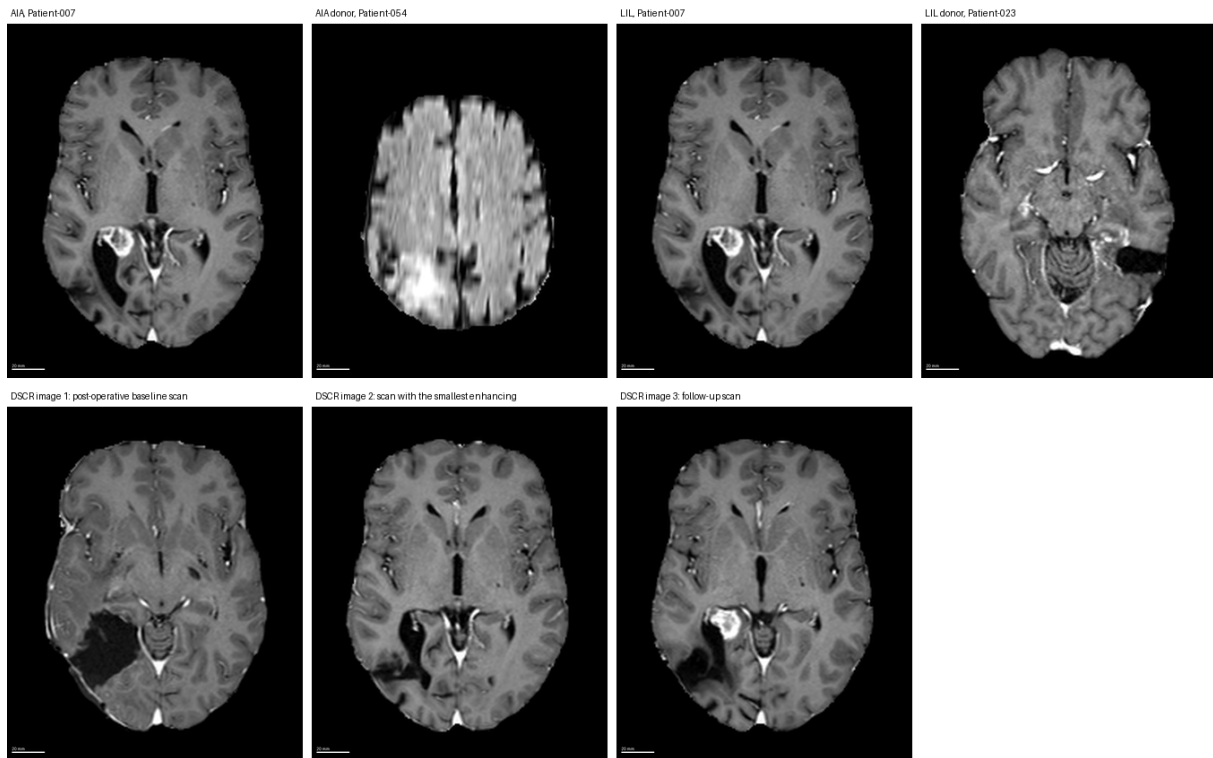


Figure 4: Images for Patient-007. Top: the AIA slice and its donor, the LIL slice and its donor. Bottom: the three DSCR slices (baseline, nadir, follow-up). Slices are atlas-space, skull-stripped, magnified $4\times$, with a 20 mm scale bar.

change therapy 20). PJRF has 45: a patient needs a recorded survival time after the scan week. A swap needs a donor with a different key and the same number of images; each patient gets one donor per phase and donates to at most two recipients, leaving 46 AIA, 38 LIL and 42 DSCR pairs (four patients have no compatible DSCR donor).

Audit size check. Where the rater recorded target-lesion diameters (81 follow-ups, 69 of them rated progressive disease), agreement was 21 of 33 where our product was within a factor of 1.5 of the rater’s and 22 of 48 otherwise, a gap this subset is too small to establish as a real difference, so it neither rules in nor rules out size discrepancy as an explanation of the disagreements; it also has few stable-disease or partial-response ratings, where disagreement concentrates, so it cannot test them.

Donor dependence (post hoc). AIA, LIL and DSCR use 31, 30 and 29 distinct donors respectively, none used more than 2 times; because donors are reused across recipients, recipients that share a donor are not independent, so a naive component bootstrap is uninformative. We instead resample recipients that share a donor together, drop each donor in turn, and permute donor keys among re-

cipients (never onto a recipient’s own key; p is the share with a gain at least as large). The permutation is a conditional label-association test and does not reconstruct the image-assignment mechanism. Draws whose repair of an own-key collision fails are discarded, so the accepted count is at most the 5,000 attempted: 5000 for the AIA cell below and 4056 for the DSCR cell (AIA and LIL reach 5,000; DSCR discards some). MedGemma-4B on AIA: gain +10.9, donor-cluster interval [+2.1, +22.7], leave-one-donor-out +6.8 to +11.4, permutation $p = .059$. Llama-4-Scout on DSCR: gain +28.6, interval [+2.5, +53.9], leave-one-donor-out +25.0 to +35.0, permutation $p = .244$. No permutation p of the twelve open-model cells is below .05.

Protocol validation with known-behaviour predictors. To illustrate that the donor-key permutation discriminates correct donor-specific tracking from a generic response to any image, we ran the same gain and permutation on synthetic predictors with known behaviour, on the real DSCR items and donor assignment (42 recipients; 50 simulations; detection rates reported with Wilson intervals against a nominal 5% level; no model is called). The permutation is a conditional label-association

test: it asks whether a recipient’s answer equals its own donor’s key more often than a key drawn from a different donor, holding the observed answers fixed and permuting donor keys among recipients (never onto a recipient’s own key); the gain is computed on the option texts, so it does not depend on the lettering, and options are already shuffled per item at load. It does not reconstruct the image-assignment mechanism (it does not preserve the equal-image-count strata or keep a reused donor’s keys together), so it licenses only that narrow comparison. A constant and a uniformly random answer give no gain and are flagged at the nominal rate. Three no-tracking predictors that respond to image presence but not to which donor is shown serve as false-positive checks: a generic image shift (the most common donor class whenever an image is present) reaches a positive gain of +19.0 points, and predictors keyed to the option position or to the number of scans shown, none flagged in more than 7% of runs (upper Wilson bound), i.e. not above the nominal rate. A genuine tracker that returns the shown donor’s key with probability ρ is flagged in 100% of runs at $\rho = 1$ and 92% at $\rho = 0.4$ (its effective probability of emitting the donor key is $\rho + (1 - \rho)/4 = 0.55$, since the guess branch can also land on it). The same separation holds on AIA and LIL (the generic shift never flagged; the noisy tracker flagged in 98% and 98% of runs) and across re-drawn valid assignments (Table 5 gives DSCR). The gain alone does not separate tracking from a generic image response, since the generic shift also has a positive gain; the permutation does, flagging the tracker but not the shift. These checks illustrate the test’s discrimination; a non-significant result for the open models indicates that we did not detect donor-specific tracking and does not rule out weaker or differently structured tracking.

Robustness to the donor assignment (post hoc). Many assignments satisfy the swap constraints, so we re-ran the models under 5 further distinct valid assignments, drawn by the same procedure, and re-computed the gain and permutation. The reference model stays strongly significant on both phases (permutation $p < .001$ in 5 of 5 AIA and 5 of 5 LIL assignments; gain +65.2 to +84.8 on AIA and +65.8 to +86.8 on LIL). Llama-4-Scout on DSCR, the one open-model cell with a positive gain, is significant in only 1 of 5 assignments (gain +19.0 to +33.3, permutation p from .039 to .706), so the non-detection does not depend on the particular

Synthetic predictor	Donor gain (pp)	Detected ($p < .05$)
Constant answer	+0.0	0%
Random answer	+2.4	0%
Generic image shift	+19.0	0%
Tracker ($\rho = 1$)	+66.7	100%
Tracker ($\rho = 0.7$)	+45.2	100%
Tracker ($\rho = 0.4$)	+19.0	92%

Table 5: Donor-protocol validation on the DSCR items (42 recipients): the donor-tracking gain and the share of 50 simulations in which the permutation flags tracking ($p < .05$), for synthetic predictors with known behaviour. A generic image shift reaches a positive gain but is not flagged; a genuine tracker is flagged even when noisy ($\rho = 0.4$).

donor assignment.

E Donor-Swap and Reference-Model Tables

Tables 6 and 7 give the donor-swap results for the open models and the reference-model results.

Precision-control detail. The two bf16/vLLM models’ image effect and donor gain (percentage points; bootstrap interval on the effect): Gemma-3-27B-bf16 is +0.0 [−10.9, +10.9] on AIA with donor gain +0.0, and +0.0 on LIL; Pixtral-12B is +4.3 [−4.3, +13.0] on AIA with donor gain −2.2, and −2.6 on LIL (its DSCR image arms are excluded; Appendix F). Every effect interval includes zero.

F Output Handling, Input Check and Fact-Flip Transitions

These are post hoc, descriptive checks. Invalid JSON has its answer letter recovered by pattern (the prompt orders it first). Of 965 records per open model (386 for the API and bf16 controls), only MedGemma-4B (105, truncation at the 800-token cap after a repeated sentence), Llama-4-Scout (46, a misspelled field name in otherwise complete output) and 1 Gemma-3-27B case produced invalid JSON; every letter was still recovered (0 and 0 unrecoverable), and Gemma-3-12B and the controls had none. The one exception is the second precision/stack control, Pixtral-12B: its DSCR image conditions returned empty output with no recoverable letter for 22 of 46 own-image and 18 of 42 swapped-image records and are excluded (its AIA and LIL arms, at most two empty per arm, are reported). MedGemma-4B’s DSCR accuracy on valid-JSON records (23.1%) stays below nominal

Model	Phase	<i>n</i>	Own	Swap: donor	Text: donor	Gain [95% CI]	Add-half CI	Exact <i>p</i>	Swap changes
MedGemma-4B	AIA	46	34.8	32.6	21.7	+10.9 [+2.2, +19.6]	[+0.9, +20.0]	.062	32.6
MedGemma-4B	LIL	38	23.7	21.1	28.9	−7.9 [−18.4, +2.6]	[−19.3, +4.3]	.375	13.2
MedGemma-4B	DSCR	42	21.4	28.6	23.8	+4.8 [−4.8, +14.3]	[−5.3, +14.4]	.625	9.5
Gemma-3-12B	AIA	46	26.1	17.4	17.4	+0.0 [−13.0, +13.0]	[−12.2, +12.2]	1.00	50.0
Gemma-3-12B	LIL	38	31.6	23.7	18.4	+5.3 [−10.5, +18.4]	[−9.6, +19.6]	.727	52.6
Gemma-3-12B	DSCR	42	54.8	35.7	31.0	+4.8 [−4.8, +14.3]	[−5.3, +14.4]	.625	2.4
Gemma-3-27B	AIA	46	32.6	34.8	30.4	+4.3 [−4.3, +13.0]	[−4.9, +13.2]	.625	32.6
Gemma-3-27B	LIL	38	23.7	26.3	21.1	+5.3 [−7.9, +18.4]	[−7.9, +17.9]	.688	44.7
Gemma-3-27B	DSCR	42	54.8	35.7	33.3	+2.4 [−4.8, +9.5]	[−6.6, +11.2]	1.00	7.1
Llama-4-Scout	AIA	46	37.0	30.4	28.3	+2.2 [−6.5, +13.0]	[−7.9, +12.1]	1.00	32.6
Llama-4-Scout	LIL	38	39.5	31.6	23.7	+7.9 [−7.9, +23.7]	[−7.8, +22.8]	.508	55.3
Llama-4-Scout	DSCR	42	54.8	50.0	21.4	+28.6 [+7.1, +50.0]	[+5.6, +49.0]	.029	47.6

Table 6: E2: share of answers (%) that equal the donor’s key when the image is swapped, and when there is no image; the gain is the difference in percentage points, with its bootstrap interval, an add-half sparse-data interval (Add-half CI) and the exact McNemar *p* (unadjusted); the last column is the share of items whose answer differs from the own-image answer after the swap. Sample sizes are the donor-paired items (AIA 46, LIL 38, DSCR 42); the own-image accuracy in Table 3 is over all items (DSCR 46).

Phase	<i>n</i>	Image	Text-only	Δ [95% CI]	Swap: donor key	Gain [95% CI]
AIA	46	95.7	21.7	+73.9 [+58.7, +87.0]	95.7	+67.4 [+54.3, +80.4]
LIL	38	100.0	36.8	+63.2 [+47.4, +78.9]	100.0	+86.8 [+76.3, +97.4]
DSCR	46	71.7	58.7	+13.0 [+0.0, +26.1]	50.0	+16.7 [+2.4, +31.0]

Table 7: Reference model (Gemini-3.6-Flash): accuracy (%) with the image and with text only, the paired difference, the share of swapped-image answers equal to the donor’s key, and the donor-tracking gain (percentage points). Accuracy is over all items (AIA 46, LIL 38, DSCR 46); the swap and donor-key columns use the donor-paired items (AIA 46, LIL 38, DSCR 42).

chance, close to its all-records score (19.6%), so parsing does not explain its below-chance score, which comes from the letters it chooses (never the fourth option). These counts do not show that images reach each model; the input check does.

Input check. For each open-model pipeline and the bf16 Gemma-3-27B rerun (the pipelines in Table 8) we sent one real AIA, one LIL and one three-image DSCR item with zero to three of its images behind a unique nonce (preventing prefix caching) and read the prompt token count. Each added image raises it by a similar amount (Table 8), so none was dropped, including the second and third DSCR images. These same pipelines passed known-answer probes (solid colours, printed numbers, image order and count, a slice’s black background), and one slice stayed legible squashed to 896 and 448 pixels. The Pixtral-12B and API-reference pipelines were not token-probed: Pixtral’s unusable DSCR image arms are excluded on output grounds above, and the API model’s internal handling cannot be inspected this way. Run once on one patient, this shows images reach the model and simple stimuli are perceived, not that MRI contrast survives each image encoder.

Table 10 gives each item’s transition under a fact flip relative to the same item under the true

context: moved to the rule’s new answer, already there, unchanged elsewhere, or changed to another answer. Only the first column shows an effect of the flip. The last column is the registered summary (moved plus already there, over all items); on the label flip it is highest for MedGemma-4B, largely because 21 of its answers were already the rule’s new answer.

PJRF (E5, secondary). The outcome’s time origin and censoring are unverified: we assume the survival time shares the week numbering’s origin, LUMIERE states neither, 4 of 86 patients have a survival time shorter than their last imaging week (1 in our cohort), and a four-week shift of the origin would flip 4 of the 45 labels. No model’s Brier score differs clearly from the constant base rate (paired difference to the leave-one-out base rate, negative better, from −0.017 to +0.009 across the four models), and a leave-one-out logistic model on age, MGMT, resection extent and weeks since surgery does not beat it either (Brier 0.283 against 0.246, AUC 0.387); MedGemma-4B and Gemma-3-12B place 78% and 91% of patients in the 0.5 bin, so their Brier is near 0.25 by construction. We draw no conclusion from PJRF (Section 3.2); this small-sample null does not show the facts carry no information.

Model	Tag	Digest	Quantization	Tokens per image	Probes passed
MedGemma-4B	medgemma:4b	9fe4e9a6c9bd	Q4_K_M	285–308	8/8
Gemma-3-12B	gemma3:12b	f4031aab637d	Q4_K_M	284–305	8/8
Gemma-3-27B	gemma3:27b	a418f5838eaf	Q4_K_M	280–304	8/8
Llama-4-Scout	llama4:scout	bf31604e25c2	Q4_K_M	1456–1477	8/8
Gemma-3-27B (bf16)	google/gemma-3-27b-it	–	bf16 (vLLM)	271–288	8/8

Table 8: Input check per pipeline: model tag, digest and quantization as reported by the server, the range of prompt-token increments per added image, and known-answer probes passed out of eight.

	TCM acc. (%)			Moved / informative	
	true	wrong	none	label (all)	timing (PD)
MedGemma-4B	54.3	39.1	34.8	10/25	1/27
Gemma-3-12B	69.6	30.4	39.1	18/40	8/21
Gemma-3-27B	69.6	32.6	45.7	25/43	0/20
Llama-4-Scout	54.3	37.0	45.7	10/34	11/21

Table 9: E3: TCM accuracy (%) under true, random wrong and absent upstream context, and the number of answers that move to the rule’s new answer after the stated category is replaced (label, all items) or the stated timing crosses the 12-week window (timing, progressive-disease items), over the informative items, those whose true-context answer was not already the new answer.

Model	Flip	Items	Moved	Already there	Unchanged	Other	Moved / informative	Registered share (%)
MedGemma-4B	label (all)	46	10	21	12	3	10/25	67.4
MedGemma-4B	timing (PD)	27	1	0	23	3	1/27	3.7
Gemma-3-12B	label (all)	46	18	6	10	12	18/40	52.2
Gemma-3-12B	timing (PD)	27	8	6	11	2	8/21	51.9
Gemma-3-27B	label (all)	46	25	3	2	16	25/43	54.3
Gemma-3-27B	timing (PD)	27	0	7	20	0	0/20	25.9
Llama-4-Scout	label (all)	46	10	12	15	9	10/34	41.3
Llama-4-Scout	timing (PD)	27	11	6	10	0	11/21	63.0

Table 10: TCM fact flips: transitions relative to the true-context answer for the same item. The label flip covers all 46 items; the timing flip covers the 27 progressive-disease items.